

system can send back one piece of data to the CPU on one access. Graphics memory on the other hand is designed to serve more than one computing core at a time. This happens due to clever organisation of memory for each processor and the way processors are grouped.

Needs and Wishes

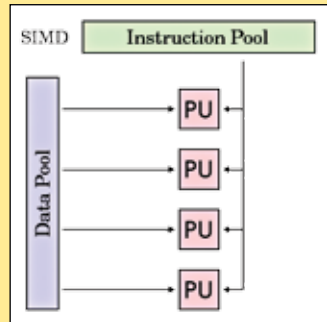
If you are not going to game a lot, skip a GPU. GPUs are built to let you play games. Multimedia editing software such as 3Ds MAX, Adobe Photoshop and After Effects too utilise GPU's powers. GPUs are power hungry devices. They can consume as much as or more power than the CPU. Make sure you are going after one only if you need one. If you are going to do multimedia editing (video editing, 3D modelling, animation or heavy image editing) or gaming, you're better off with a dedicated GPU. Almost all other requirements can be fulfilled by on-chip GPUs built into processors.

That said, most mainstream laptops come with a GPU and you are out of choices. If you are planning to buy a laptop from the mainstream segment (in the ₹30K – ₹50K price range), look for 'switchable' graphics. Such machines utilize the processor's on-chip graphics for most cases and use GPUs only for graphics-heavy jobs which cannot be served well by the on-chip GPUs.

You might wonder why we left HD video playback out of the list of GPU-intensive jobs. It's because only a few media players such as Windows Media Player and iTunes utilise the powers of GPU for video playback while the

NOTE: What's SIMD?

SIMD stands for Single Instruction Multiple Data and is a computer architecture which allows fast calculation on an array of similar data. A GPU is



a SIMD type system while a modern CPU is Single Instruction Single Data (SISD) processor with Symmetric Multiprocessing (SMP) (they come close to MIMD architecture). A CPU runs multiple threads each of which runs on its own. They all can run different instructions in the same clock cycles, e.g. while one is trying to do multiplication, second one can be doing subtraction, a third one could be trying to access memory and so on and so forth.